Lab NO: 01

Lab name: Desingning a calculator with basic Mathematical operation using GUI component.

Theory: In Java, GUI (Graphical User Interface) applications are commonly developed using the Swing or AWT frameworks. These libraries provide components such as buttons, text fields and panels to design user friendly interfaces.

A basic calculator typically includes buttons for digits (0-9) operators (+, -, X, ÷) and functions like clear and equals. Event-driven programming plays a critical role here, Each button is associated with an event listener that defines its behavior when clicked.

Key GUI Components used:

- JFrame — the main window container
- JPanel — to organiz components.
- JButton — for operations and digits.
- JTextField — to display input and results.

The calculator logic involves capturing button press, parsing numbers, and performing operations accordingly. Exception handling is also important to manage input errors (e.g. divide by zero).

Lab No: 02

Lab Name: Java program to convert binary to decimal and decimal to binary using GUI.

Theory: Binary numbers consist of only 0s and 1s and are commonly used in digital systems, while decimal numbers are the standard base-10 numbers used in everyday arithmetic.

Number base conversion involves changing a number from one numerical base to another. In this case

- Decimal to Binary: This conversion is typically done using repeated division by 2 and collecting the remainders.

- Binary to Decimal: This involves computing the sum of each binary digit (bit) multiplied by 2 raised to the position's power.

Using Java, input from the user is taken through a JTextField, and the converted result is displayed using another JTextField or JLabel.

Lab No: 03

Lab Name: 2-player Tie-Toe-Toe using Java GUI

Theory: A 2-player Tie-Toe-Toe game is implemented using Swing library. The game consists of a 3X3 grid where two players taken turns marking spaces with 'X' and "O". The interface is created using components such as JFrame for the window and JButton for each cell of the grid.

A Grid Layout manager is used to arrange the button in a uniform 3X3 layout. Each Button respond to user input through an ActionListener, which updates the button label with the current player's ~~symbol~~ symbol and disables further interaction with the cell.

Game logic checks for winning conditions after each move by evaluating all possible row, column, and ~~dig~~ diagonal combinations.

Lab No: 04

Lab Name: Java program to illustrate suspend, resume and stop operation of thread.

Theory: Threads allow a Java program to perform multiple tasks concurrently. A thread represents a single path of execution within a program. Java provides built-in support for multithreading through the Thread class and the Runnable interface. Although the ~~Ag~~ suspend (), resume (), and stop() methods of the Thread class were originally introduced to control thread behavior, they are now deprecated due to safety issues like deadlocks and inconsistent thread states. However they are still studied for understanding basic thread control mechanisms.

- suspend (): Temporarily halts the execution of a thread, keeping it in memory.
- resume (): Restarts a suspended thread.
- stop (): Immediately terminates a thread's execution.

Lab NO: 05

Lab Name: Writing a Java program to create thread class.

Theory: Java supports multithreading allowing multiple sequences of execution to run in parallel. A Thread in Java can be created in two primary ways:

1. Extending the Thread class
2. Implementing the Runnable interface.

Extending Thread class involves:

(i) Defining a new class that inherits from Thread.
(ii) Overriding the run() method to specify the task that the thread should perform.
(iii) Creating an instance of the thread class and calling start() to begin execution.

Once start() is called, a new thread is created and the run() method is executed concurrently with the main program. The start() method ensures that the thread runs in separate call stack.

• stop() : Immediately terminat...

Lab No: 06

Lab Name: Java program to illustrate yield(), stop() and sleep() method using thread.

Theory: Thread control in Java can be fine tuned using several built-in methods that influence how and when threads execute. ~~In thread~~

- sleep (milliseconds): Temporarily pauses the execution of the current thread for a specified amount of time. This does not release any locks the thread may hold.
- yield(): Suggests to the thread scheduler that current thread is willing to pause and lets other threads of equal priority execute. It does not guarantee that the current thread will be paused.
- stop(): Immediately terminates the thread. This method is deprecated because it can leave shared resources in inconsistent state and is unsafe in most cases.

Lab No: 07

Lab Name: Client and server program in Java to establish a connection between them.

Theory: Networking in Java is supported through the java.net package, which allows communication between programs over a network using sockets. A socket provides an endpoint for sending and receiving data. This demonstration of how to establish a basic connection between a client and a server using TCP is:

- The server uses a ServerSocket to listen for incoming client connections on a specified port.
- The client uses a Socket to initiate a connection to the server using the server's IP address and port.
- Once the connection is established, both client and server communicate by sending and receiving data through input and output streams.

Lab No : 08

Lab Name: Designing a form with Java GUI to perform insert, delete, edit, and show data using Excel database.

Theory: A data entry form is created using Java's GUI framework to perform CRUD operations - Create, Read, Update, Delete - on a dataset stored in a Excel file. The form simulates a basic database interface, allowing users to manage data records through a graphical interface.

The GUI is built with components such as:

- JText field : for input fields.
- JButton : to trigger insert, delete, edit, or display actions.
- JTable : to show existing data in a tabular format.
- JLabel : to label each input field.

To handle Excel files, the application uses external libraryies like Apache POI, which enables Java programs to read from and write to Excel files.